

B.E. ELECTRICAL ENGINEERING
SECOND YEAR
SECOND SEMESTER EXAM 2023
FIELD THEORY
PART-II

Solved Answer Script in Academic Article Style

Time: 3 hours

Use separate answer-script for each part

Full Marks: 100

50 marks for each part

Answer any three questions. Two marks are for neatness. All symbols have their usual significance.

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1. (a) What is the definition of curl? Derive the expression for curl of a vector field $\mathbf{M}$ using definition of curl, written as $\nabla \times \mathbf{M}$ where ∇ is the vector differential operator in Cartesian co-ordinates.

Short answer-building plan

First define curl as circulation per unit area. Then take a small rectangular loop in the xy -plane. Find the line integral of $\mathbf{M}$ around the four sides. Divide by the area $dx dy$ to get the z -component. Then write the complete Cartesian expression by cyclic symmetry.

Detailed semester-style answer

Curl of a vector field gives the local rotational tendency or circulation tendency of the field at a point.

For a vector field $\mathbf{M}$, the component of curl along unit normal $\hat{\mathbf{n}}$ is defined as

$$(\nabla \times \mathbf{M}) \cdot \hat{\mathbf{n}} = \lim_{\Delta S \rightarrow 0} \frac{1}{\Delta S} \oint_C \mathbf{M} \cdot d\mathbf{l}$$

where C is the closed boundary of the small surface ΔS .

Let

$$\mathbf{M} = M_x \hat{\mathbf{x}} + M_y \hat{\mathbf{y}} + M_z \hat{\mathbf{z}}.$$

To derive the z -component of curl, consider a small rectangular path in the xy -plane.

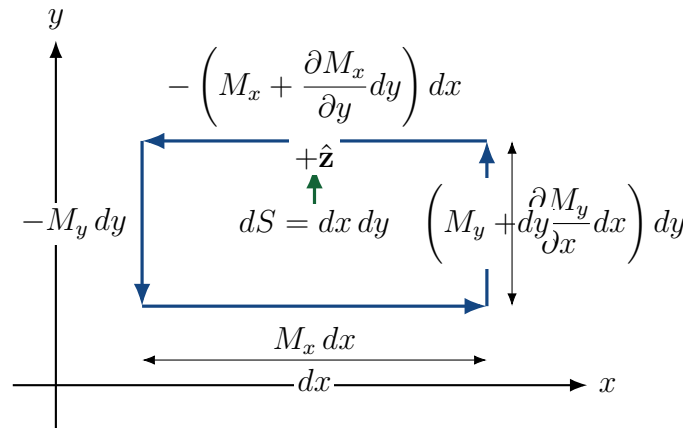


Figure 1: Small rectangular path in the xy -plane for deriving the z -component of curl.

Detailed semester-style answer

The circulation around the four sides is

$$\oint \mathbf{M} \cdot d\mathbf{l} = \int_1 + \int_2 + \int_3 + \int_4.$$

For side 1:

$$\int_1 \mathbf{M} \cdot d\mathbf{l} = M_x dx.$$

For side 2:

$$\int_2 \mathbf{M} \cdot d\mathbf{l} = \left(M_y + \frac{\partial M_y}{\partial x} dx \right) dy.$$

For side 3:

$$\int_3 \mathbf{M} \cdot d\mathbf{l} = - \left(M_x + \frac{\partial M_x}{\partial y} dy \right) dx.$$

For side 4:

$$\int_4 \mathbf{M} \cdot d\mathbf{l} = -M_y dy.$$

Adding,

$$\begin{aligned} \oint \mathbf{M} \cdot d\mathbf{l} &= M_x dx + \left(M_y + \frac{\partial M_y}{\partial x} dx \right) dy \\ &\quad - \left(M_x + \frac{\partial M_x}{\partial y} dy \right) dx - M_y dy. \end{aligned}$$

After cancellation,

$$\oint \mathbf{M} \cdot d\mathbf{l} = \left(\frac{\partial M_y}{\partial x} - \frac{\partial M_x}{\partial y} \right) dx dy.$$

Therefore,

$$(\nabla \times \mathbf{M})_z = \frac{\partial M_y}{\partial x} - \frac{\partial M_x}{\partial y}$$

Similarly,

$$(\nabla \times \mathbf{M})_x = \frac{\partial M_z}{\partial y} - \frac{\partial M_y}{\partial z}$$

and

$$(\nabla \times \mathbf{M})_y = \frac{\partial M_x}{\partial z} - \frac{\partial M_z}{\partial x}$$

Final result / exam point

Thus,

$$\nabla \times \mathbf{M} = \left(\frac{\partial M_z}{\partial y} - \frac{\partial M_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial M_x}{\partial z} - \frac{\partial M_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial M_y}{\partial x} - \frac{\partial M_x}{\partial y} \right) \hat{\mathbf{z}}$$

In determinant form:

$$\nabla \times \mathbf{M} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ M_x & M_y & M_z \end{vmatrix}$$

1. (b) Establish Boundary Conditions for magnetic field when it passes through two different magnetic media having permeabilities μ_1 and μ_2 respectively.

Short answer-building plan

Draw the interface between two magnetic media. Use Ampere's law for a small rectangular loop to get the tangential component condition of $\mathbf{H}$. Use Gauss's law for magnetism for a small pillbox to get the normal component condition of $\mathbf{B}$. Then combine the two conditions.

Detailed semester-style answer

Consider two magnetic media separated by a plane boundary.

Let the permeabilities be

$$\mu_1 \quad \text{and} \quad \mu_2.$$

The magnetic flux density and magnetic field intensity are related by

$$\mathbf{B}_1 = \mu_1 \mathbf{H}_1, \quad \mathbf{B}_2 = \mu_2 \mathbf{H}_2$$

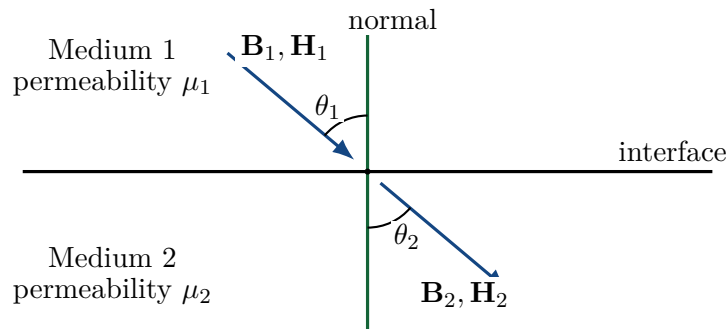


Figure 2: Magnetic field passing through two magnetic media.

Detailed semester-style answer

Condition 1: Tangential component of $\mathbf{H}$

Apply Ampere's circuital law to a very small rectangular loop crossing the boundary:

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}}.$$

If there is no surface current at the boundary,

$$I_{\text{enc}} = 0.$$

Therefore,

$$H_{1t}\Delta l - H_{2t}\Delta l = 0.$$

Hence,

$$H_{1t} = H_{2t}$$

If angles are measured from the normal, then

$$H_1 \sin \theta_1 = H_2 \sin \theta_2$$

Condition 2: Normal component of B

Gauss's law for magnetism is

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0$$

Apply it to a small pillbox across the boundary:

$$B_{1n}\Delta S - B_{2n}\Delta S = 0.$$

Therefore,

$$B_{1n} = B_{2n}$$

If angles are measured from the normal:

$$B_1 \cos \theta_1 = B_2 \cos \theta_2$$

Using

$$B_1 = \mu_1 H_1, \quad B_2 = \mu_2 H_2,$$

we get

$$\mu_1 H_1 \cos \theta_1 = \mu_2 H_2 \cos \theta_2.$$

Combining with

$$H_1 \sin \theta_1 = H_2 \sin \theta_2,$$

we obtain the magnetic refraction relation:

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\mu_1}{\mu_2}$$

Final result / exam point

Boundary conditions for magnetic field are:

$$H_{1t} = H_{2t}$$

and

$$B_{1n} = B_{2n}$$

Also,

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\mu_1}{\mu_2}$$

2. (a) Establish the Stoke's Theorem.

Short answer-building plan

First state Stoke's theorem. Then explain it by dividing an open surface into many small surface elements. Internal boundaries cancel in pairs, and only the outer boundary remains.

Detailed semester-style answer

Stoke's theorem states that the line integral of a vector field around a closed boundary is equal to the surface integral of the curl of that vector field over the open surface bounded by that boundary.

Mathematically,

$$\oint_C \mathbf{A} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

Here:

- C is the closed boundary curve,
- S is the open surface bounded by C ,
- $d\mathbf{S}$ is the vector surface element.

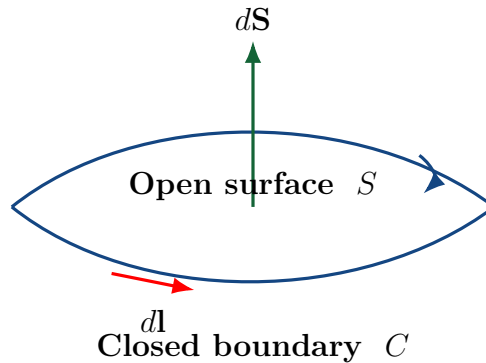


Figure 3: Stoke's theorem connects circulation along boundary with curl over surface.

Detailed semester-style answer

For a small element of the surface,

$$(\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

represents the circulation of $\mathbf{A}$ around the boundary of that small element.

Now divide the complete surface S into many small elements. When all small circulations are added, the internal boundary line integrals cancel because they are taken in opposite directions. Only the outer boundary C remains.

Therefore,

$$\sum (\nabla \times \mathbf{A}) \cdot d\mathbf{S} = \oint_C \mathbf{A} \cdot d\mathbf{l}.$$

In the limit,

$$\iint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S} = \oint_C \mathbf{A} \cdot d\mathbf{l}$$

Thus Stoke's theorem is established.

2. (b) Derive the expression for Divergence of $\mathbf{J}$ and show $\nabla \cdot \mathbf{J} = 0$.

Short answer-building plan

First define divergence as outward flux per unit volume. Then derive the Cartesian expression for $\nabla \cdot \mathbf{J}$. Finally use steady current condition or Ampere's law to show $\nabla \cdot \mathbf{J} = 0$.

Detailed semester-style answer

Divergence of current density $\mathbf{J}$ is the net outward current per unit volume.

By fundamental definition,

$$\nabla \cdot \mathbf{J} = \lim_{\Delta V \rightarrow 0} \frac{1}{\Delta V} \oint_S \mathbf{J} \cdot d\mathbf{S}$$

Let

$$\mathbf{J} = J_x \hat{\mathbf{x}} + J_y \hat{\mathbf{y}} + J_z \hat{\mathbf{z}}.$$

For a small rectangular box of volume

$$\Delta V = dx \, dy \, dz,$$

the net outward current through opposite faces is:

$$\oint_S \mathbf{J} \cdot d\mathbf{S} = \left(\frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} + \frac{\partial J_z}{\partial z} \right) dx \, dy \, dz.$$

Therefore,

$$\nabla \cdot \mathbf{J} = \frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} + \frac{\partial J_z}{\partial z}$$

Detailed semester-style answer

Now from Ampere's law,

$$\nabla \times \mathbf{H} = \mathbf{J}$$

Taking divergence on both sides:

$$\nabla \cdot (\nabla \times \mathbf{H}) = \nabla \cdot \mathbf{J}.$$

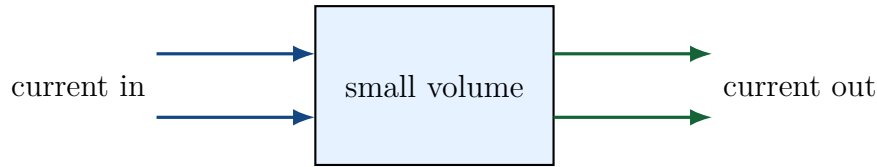
But for any vector field,

$$\nabla \cdot (\nabla \times \mathbf{H}) = 0$$

Therefore,

$$\nabla \cdot \mathbf{J} = 0$$

This is valid for steady current distribution.



$$\nabla \cdot \mathbf{J} = 0: \text{no charge accumulation}$$

Figure 4: For steady current, current entering equals current leaving.

Final result / exam point

$$\nabla \cdot \mathbf{J} = 0$$

means no charge accumulation occurs inside the volume. It is the field form of Kirchhoff's current law for steady current.

2. (c) Establish $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$ and show the pictorial view of this relation.

Short answer-building plan

Start from Faraday's law in integral form. Use magnetic flux through the surface. Apply Stoke's theorem to convert the electric field line integral into surface integral of curl. Then compare the integrands.

Detailed semester-style answer

Faraday's law states that a time-varying magnetic flux produces an induced electric field. The integral form is

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \iint_S \mathbf{B} \cdot d\mathbf{S}$$

For a stationary surface,

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = -\iint_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}$$

Using Stoke's theorem,

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S}.$$

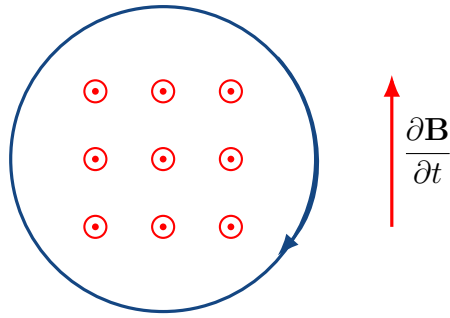
Thus,

$$\iint_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S} = -\iint_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}.$$

Since the surface is arbitrary,

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Changing magnetic field



Induced circulating electric field $\mathbf{E}$

Figure 5: Time-varying magnetic field produces circulating electric field.

Final result / exam point

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

The negative sign represents Lenz's law.

3. (a) Establish $\nabla \times \mathbf{H} = \mathbf{J}$ and explain its physical significance.

Short answer-building plan

Start from Ampere's circuital law. Write enclosed current as surface integral of current density. Use Stoke's theorem to convert the line integral into curl form. Then explain that current density creates circulation of magnetic field intensity.

Detailed semester-style answer

Ampere's circuital law states that

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}}$$

The current enclosed by the contour can be written as

$$I_{\text{enc}} = \iint_S \mathbf{J} \cdot d\mathbf{S}$$

Therefore,

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = \iint_S \mathbf{J} \cdot d\mathbf{S}.$$

Using Stoke's theorem,

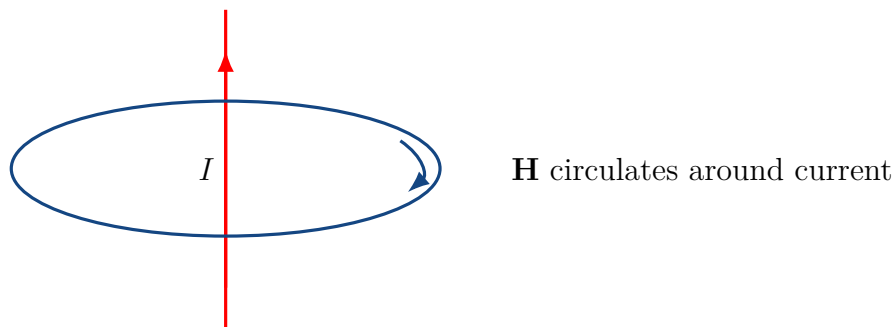
$$\oint_C \mathbf{H} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S}.$$

Hence,

$$\iint_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S} = \iint_S \mathbf{J} \cdot d\mathbf{S}.$$

Since the surface is arbitrary,

$$\nabla \times \mathbf{H} = \mathbf{J}$$



Current density $\mathbf{J}$ pierces the surface

Figure 6: Physical significance of $\nabla \times \mathbf{H} = \mathbf{J}$.

Final result / exam point

$$\nabla \times \mathbf{H} = \mathbf{J}$$

means current density is the source of magnetic field circulation.

If $\mathbf{J} = 0$, then in that region

$$\nabla \times \mathbf{H} = 0.$$

So magnetic field intensity has no local circulation in a current-free region.

3. (b) A square loop measuring 2 m by 2 m carries a 7.5 A steady current as shown in figure 1, where the loop is in the xz plane, the origin coincide with a corner of the square. Using Biot-Savart law compute the B-field at a point on y -axis 0.35 m from the origin in air material.

Short answer-building plan

Place the square loop in the xz -plane with one corner at the origin. Take the point P on the y -axis at $d = 0.35\text{ m}$. Use Biot-Savart law for each of the four straight sides. Add the four vector contributions to get the total magnetic flux density.

Detailed semester-style answer

Given:

$$L = 2\text{ m}, \quad I = 7.5\text{ A}, \quad d = 0.35\text{ m}.$$

Let the square loop lie in the xz -plane. Let one corner be at the origin $O(0, 0, 0)$ and the field point be

$$P(0, d, 0).$$

For the current direction shown in the diagram, the direction of $\mathbf{B}$ is obtained from the right-hand rule. If current direction is reversed, the sign of $\mathbf{B}$ will reverse, but magnitude will remain the same.

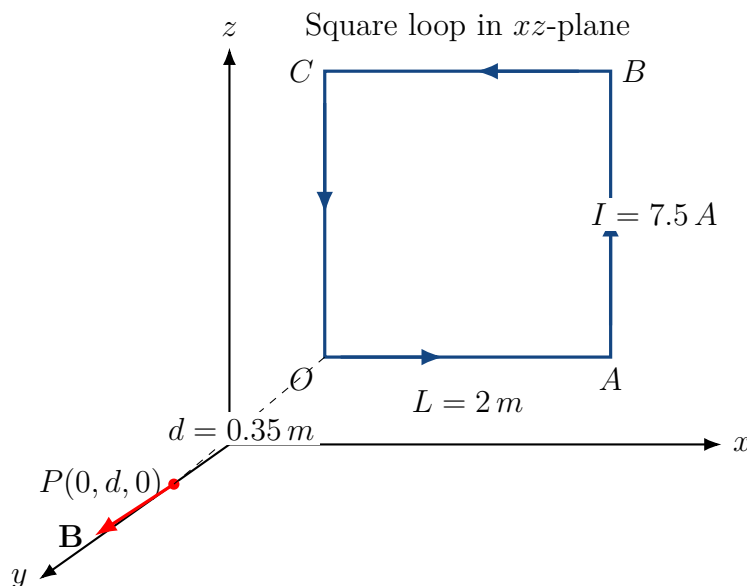


Figure 7: Square current loop with one corner at origin and point P on the y -axis.

Detailed semester-style answer

Biot-Savart law is

$$d\mathbf{B} = \frac{\mu_0 I}{4\pi} \frac{d\mathbf{l} \times \mathbf{R}}{R^3}$$

where $\mathbf{R}$ is the vector from current element to the field point.

For the four sides of the square, after integration, the total magnetic field can be written as

$$\mathbf{B} = \frac{\mu_0 I}{4\pi} [(-K_1 + dK_2)\hat{\mathbf{x}} + (2LK_2)\hat{\mathbf{y}} + (K_1 - dK_2)\hat{\mathbf{z}}]$$

where

$$K_1 = \frac{L}{d\sqrt{L^2 + d^2}}$$

and

$$K_2 = \frac{L}{(L^2 + d^2)\sqrt{2L^2 + d^2}}$$

Now,

$$L = 2\text{ m}, \quad d = 0.35\text{ m}.$$

$$K_1 = \frac{2}{0.35\sqrt{2^2 + 0.35^2}}$$

$$K_1 = \frac{2}{0.35\sqrt{4.1225}} \approx 2.8144.$$

Also,

$$K_2 = \frac{2}{(4.1225)\sqrt{8.1225}} \approx 0.1702.$$

Since

$$\frac{\mu_0 I}{4\pi} = \frac{(4\pi \times 10^{-7})(7.5)}{4\pi} = 7.5 \times 10^{-7},$$

we get

$$B_x = (7.5 \times 10^{-7})(-2.8144 + 0.35 \times 0.1702)$$

$$B_x \approx -2.066 \times 10^{-6} \text{ T}$$

$$B_y = (7.5 \times 10^{-7})(2 \times 2 \times 0.1702)$$

$$B_y \approx 5.107 \times 10^{-7} \text{ T}$$

$$B_z = (7.5 \times 10^{-7})(2.8144 - 0.35 \times 0.1702)$$

$$B_z \approx 2.066 \times 10^{-6} \text{ T}$$

Therefore,

$$\mathbf{B} = -2.066 \times 10^{-6} \hat{\mathbf{x}} + 5.107 \times 10^{-7} \hat{\mathbf{y}} + 2.066 \times 10^{-6} \hat{\mathbf{z}} \text{ T}$$

Magnitude:

$$|\mathbf{B}| = \sqrt{B_x^2 + B_y^2 + B_z^2}$$

$$|\mathbf{B}| \approx 2.97 \times 10^{-6} T$$

Final result / exam point

For the assumed current direction:

$$\mathbf{B} = -2.066\hat{\mathbf{x}} + 0.511\hat{\mathbf{y}} + 2.066\hat{\mathbf{z}} \quad \mu T$$

$$|\mathbf{B}| \approx 2.97 \mu T$$

If current direction is reversed, the direction of $\mathbf{B}$ reverses.

4. (a) Derive Electromagnetic Wave equations using Maxwell's equations.

Short answer-building plan

Write Maxwell's curl equations in a homogeneous conducting medium. Use $\mathbf{D} = \epsilon\mathbf{E}$, $\mathbf{B} = \mu\mathbf{H}$ and $\mathbf{J} = \sigma\mathbf{E}$. Take curl of Faraday's law to get wave equation for $\mathbf{E}$. Take curl of Ampere-Maxwell law to get wave equation for $\mathbf{H}$.

Detailed semester-style answer

For a homogeneous conducting medium:

$$\mathbf{D} = \epsilon\mathbf{E}, \quad \mathbf{B} = \mu\mathbf{H}, \quad \mathbf{J} = \sigma\mathbf{E}.$$

Maxwell's curl equations are:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} = -\mu \frac{\partial \mathbf{H}}{\partial t}$$

and

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} = \sigma\mathbf{E} + \epsilon \frac{\partial \mathbf{E}}{\partial t}$$

Wave equation for electric field:

Take curl of Faraday's law:

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu \frac{\partial}{\partial t} (\nabla \times \mathbf{H}).$$

Substitute:

$$\nabla \times \mathbf{H} = \sigma\mathbf{E} + \epsilon \frac{\partial \mathbf{E}}{\partial t}.$$

Then

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu\sigma \frac{\partial \mathbf{E}}{\partial t} - \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}.$$

Using vector identity:

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}.$$

In source-free region,

$$\nabla \cdot \mathbf{E} = 0.$$

So,

$$-\nabla^2 \mathbf{E} = -\mu\sigma \frac{\partial \mathbf{E}}{\partial t} - \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}.$$

Therefore,

$$\nabla^2 \mathbf{E} = \mu\sigma \frac{\partial \mathbf{E}}{\partial t} + \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

Similarly, the magnetic field wave equation is

$$\nabla^2 \mathbf{H} = \mu\sigma \frac{\partial \mathbf{H}}{\partial t} + \mu\epsilon \frac{\partial^2 \mathbf{H}}{\partial t^2}$$

Final result / exam point

Electromagnetic wave equations are:

$$\nabla^2 \mathbf{E} = \mu\sigma \frac{\partial \mathbf{E}}{\partial t} + \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

$$\nabla^2 \mathbf{H} = \mu\sigma \frac{\partial \mathbf{H}}{\partial t} + \mu\epsilon \frac{\partial^2 \mathbf{H}}{\partial t^2}$$

4. (b) What is plane electromagnetic wave? Using electromagnetic wave equation of Magnetic Field ($\nabla^2 \mathbf{H} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{H}}{\partial t^2}$) in free space, obtain an analytical solution of the wave equation of Magnetic Field ($\mathbf{H}$) considering it as a plane wave and also draw the wave propagation.

Short answer-building plan

Define plane electromagnetic wave first. Then write the free-space magnetic field wave equation. Assume propagation along $+x$ direction and magnetic field along z direction. Reduce the vector equation to scalar form and write sinusoidal travelling wave solution.

Detailed semester-style answer

A plane electromagnetic wave is a wave in which the field quantities are constant over any plane perpendicular to the direction of propagation.

In a plane EM wave:

$$\mathbf{E} \perp \mathbf{H}, \quad \mathbf{E} \perp \text{direction of propagation}, \quad \mathbf{H} \perp \text{direction of propagation}.$$

Also,

$$\mathbf{E} \times \mathbf{H}$$

gives the direction of propagation.

In free space,

$$\nabla^2 \mathbf{H} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{H}}{\partial t^2}$$

Let the wave travel along the $+x$ direction and let

$$\mathbf{H} = H_z(x, t) \hat{\mathbf{z}}$$

Since it is a plane wave travelling along x ,

$$\frac{\partial}{\partial y} = 0, \quad \frac{\partial}{\partial z} = 0.$$

Therefore,

$$\nabla^2 \mathbf{H} = \frac{\partial^2 H_z}{\partial x^2} \hat{\mathbf{z}}.$$

So,

$$\frac{\partial^2 H_z}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 H_z}{\partial t^2}$$

A sinusoidal travelling wave solution is

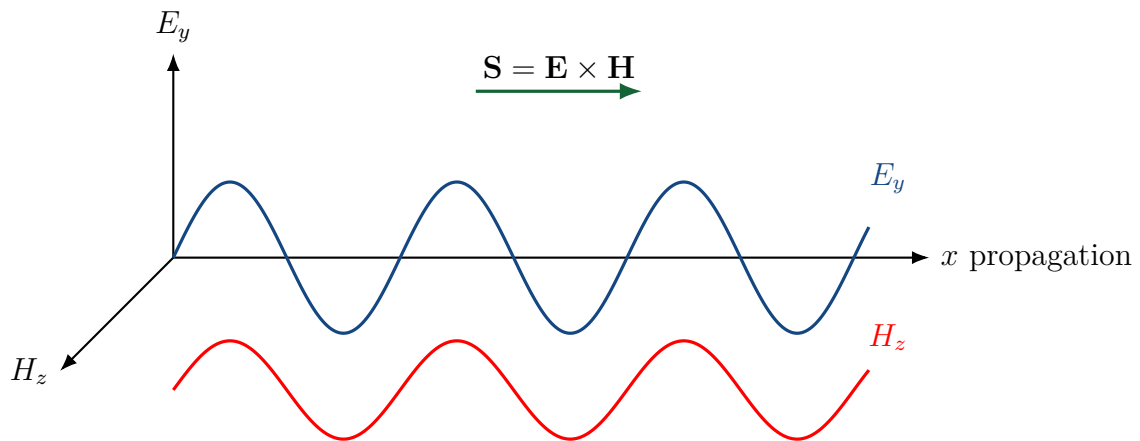
$$H_z(x, t) = H_0 \cos(\omega t - \beta x)$$

where

$$\beta = \omega \sqrt{\mu_0 \epsilon_0}$$

The velocity is

$$v = \frac{\omega}{\beta} = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$



Plane EM wave: $\mathbf{E}$, $\mathbf{H}$ and propagation direction are mutually perpendicular

Figure 8: Plane electromagnetic wave propagation in free space.

Final result / exam point

$$H_z(x, t) = H_0 \cos(\omega t - \beta x)$$

$$\beta = \omega \sqrt{\mu_0 \epsilon_0}, \quad v = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

In free space:

$$v = c \approx 3 \times 10^8 \text{ m/s}$$

5. Write short notes on any two.

Short answer-building plan

The paper asks for any two short notes. All three are solved here. In the exam, select the two you can present most confidently.

5. (a) $\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$ and the pictorial view of this relation.

Detailed semester-style answer

For steady fields, Ampere's law is

$$\nabla \times \mathbf{H} = \mathbf{J}.$$

For time-varying fields, Maxwell added displacement current density:

$$\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t}$$

Therefore, the complete Ampere-Maxwell law is

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

Here:

- $\mathbf{J}$ is conduction current density,
- $\frac{\partial \mathbf{D}}{\partial t}$ is displacement current density.

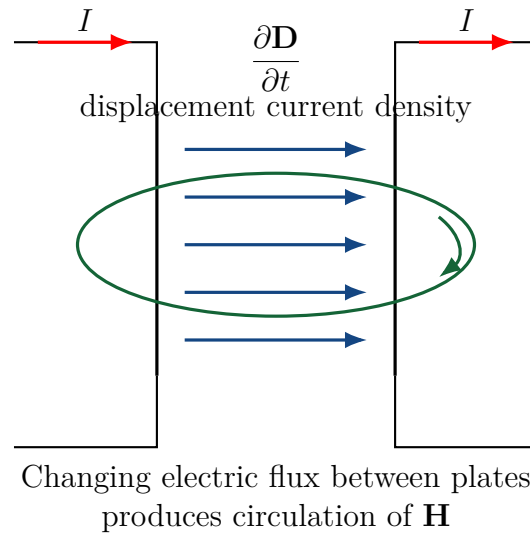


Figure 9: Pictorial view of Ampere-Maxwell law.

Final result / exam point

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

means both conduction current and changing electric flux produce circulation of magnetic field intensity.

5. (b) Electromagnetic wave polarizations: Linear, Circular, Elliptical.

Detailed semester-style answer

Polarization of an electromagnetic wave describes the manner in which the electric field vector $\mathbf{E}$ varies with time at a fixed point in space.

The path traced by the tip of the electric field vector determines the type of polarization.

Path traced by tip of electric field vector

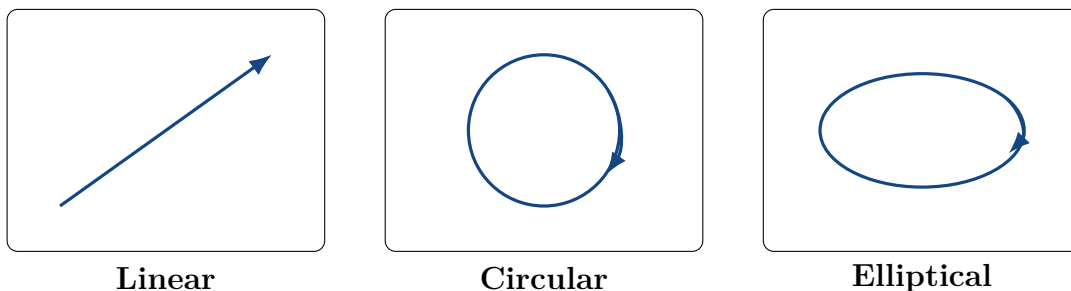


Figure 10: Linear, circular and elliptical polarizations.

Detailed semester-style answer

Linear polarization:

If the electric field vector always remains along a fixed straight line, the wave is linearly polarized.

$$\mathbf{E} = E_0 \cos(\omega t - \beta z) \hat{\mathbf{x}}$$

Circular polarization:

If two perpendicular electric field components have equal amplitude and phase difference of 90° , then the tip of $\mathbf{E}$ traces a circle.

Condition:

$$E_{0x} = E_{0y}, \quad \delta = \pm \frac{\pi}{2}$$

Elliptical polarization:

If the two perpendicular electric field components have unequal amplitudes or any general phase difference, the tip of $\mathbf{E}$ traces an ellipse.

Elliptical polarization is the most general case. Linear and circular polarizations are special cases.

Final result / exam point

Polarization is decided by the time variation of electric field vector.

The three common types are:

Linear, Circular and Elliptical

5. (c) Poynting Theorem.

Detailed semester-style answer

Poynting vector represents electromagnetic power flow per unit area.

It is defined as

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}$$

Its unit is

$$\text{W/m}^2$$

Poynting theorem states that the net electromagnetic power flowing out of a volume is equal to the rate of decrease of stored electromagnetic energy minus the ohmic loss.

Detailed semester-style answer

Electromagnetic energy density is

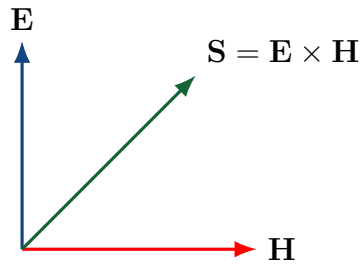
$$u_{em} = \frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2.$$

The differential form of Poynting theorem is

$$\nabla \cdot \mathbf{S} = -\frac{\partial u_{em}}{\partial t} - \mathbf{E} \cdot \mathbf{J}$$

The integral form is

$$\oint_S \mathbf{S} \cdot d\mathbf{S} = -\frac{\partial}{\partial t} \int_V \left(\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right) dV - \int_V \mathbf{E} \cdot \mathbf{J} dV$$



Direction of energy flow

Figure 11: Poynting vector gives electromagnetic energy flow direction.

Final result / exam point

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}$$

Power leaving volume = rate of decrease of stored EM energy – ohmic loss